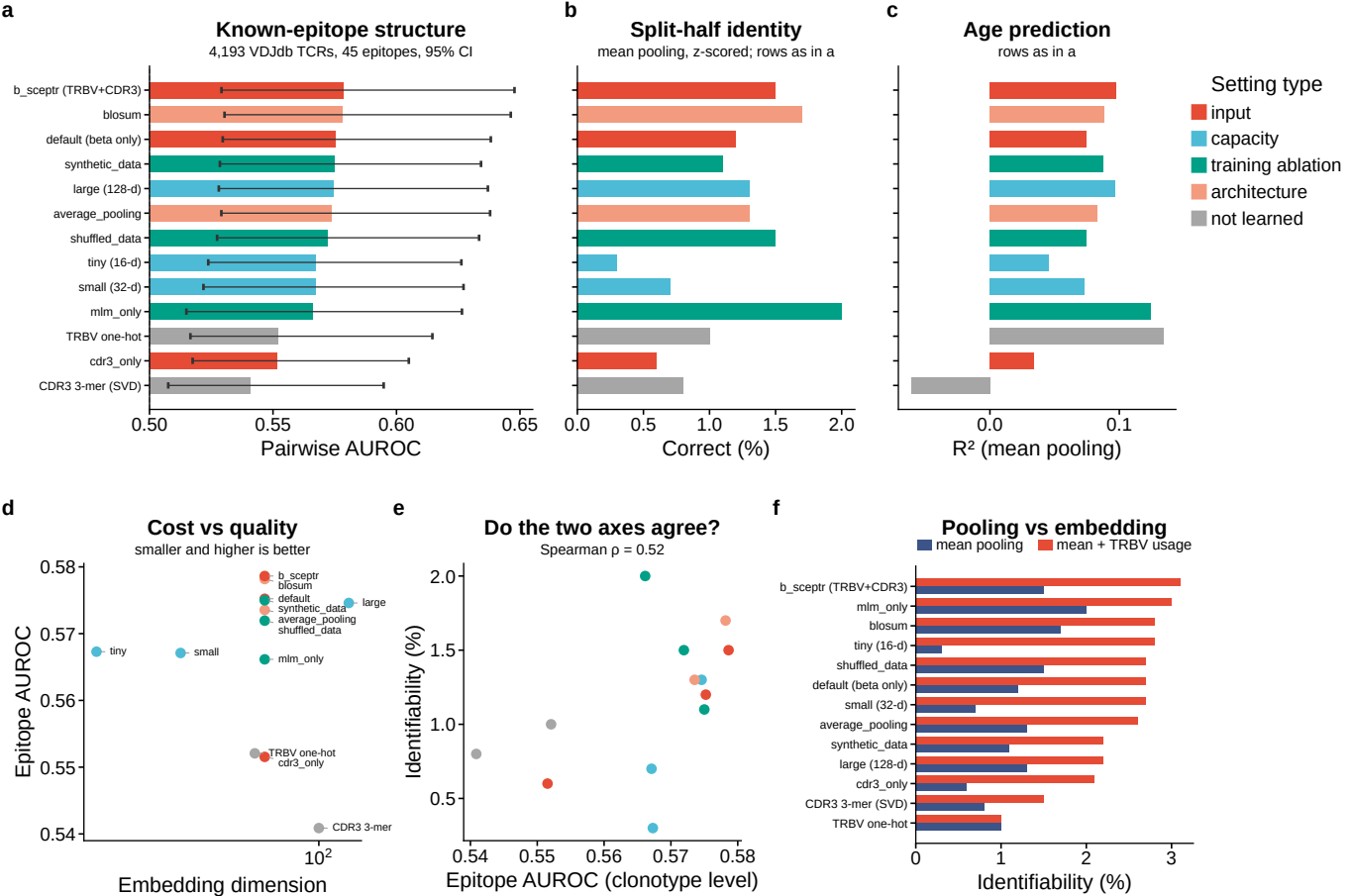
a
Known-epitope structure
4,193 VDJdb TCRs, 45 epitopes, 95% CI
b_sceptr (TRBV+CDR3)
blosum
default (beta only)
synthetic_data
large (128-d)
average_pooling
shuffled_data
tiny (16-d)
small (32-d)
mlm_only
TRBV one-hot
cdr3_only
CDR3 3-mer (SVD)
0.50
0.55
0.60
0.65
Pairwise AUROC
b
Split-half identity
mean pooling, z-scored; rows as in a
0.0
0.5
1.0
1.5
2.0
Correct (%)
c
Age prediction
rows as in a
0.0
0.1
R² (mean pooling)
Setting type
input
capacity
training ablation
architecture
not learned
d
Cost vs quality
smaller and higher is better
0.58
0.57
0.56
0.55
0.54
Epitope AUROC
b_sceptr
blosum
default
synthetic_data
average_pooling
shuffled_data
large
tiny
small
mlm_only
TRBV one-hot
cdr3_only
CDR3 3-mer
10²
Embedding dimension
e
Do the two axes agree?
Spearman ρ = 0.52
2.0
1.5
1.0
0.5
Identifiability (%)
0.54
0.55
0.56
0.57
0.58
Epitope AUROC (clonotype level)
f
Pooling vs embedding
mean pooling
mean + TRBV usage
b_sceptr (TRBV+CDR3)
mlm_only
blosum
tiny (16-d)
shuffled_data
default (beta only)
small (32-d)
average_pooling
synthetic_data
large (128-d)
cdr3_only
CDR3 3-mer (SVD)
TRBV one-hot
0
1
2
3
Identifiability (%)